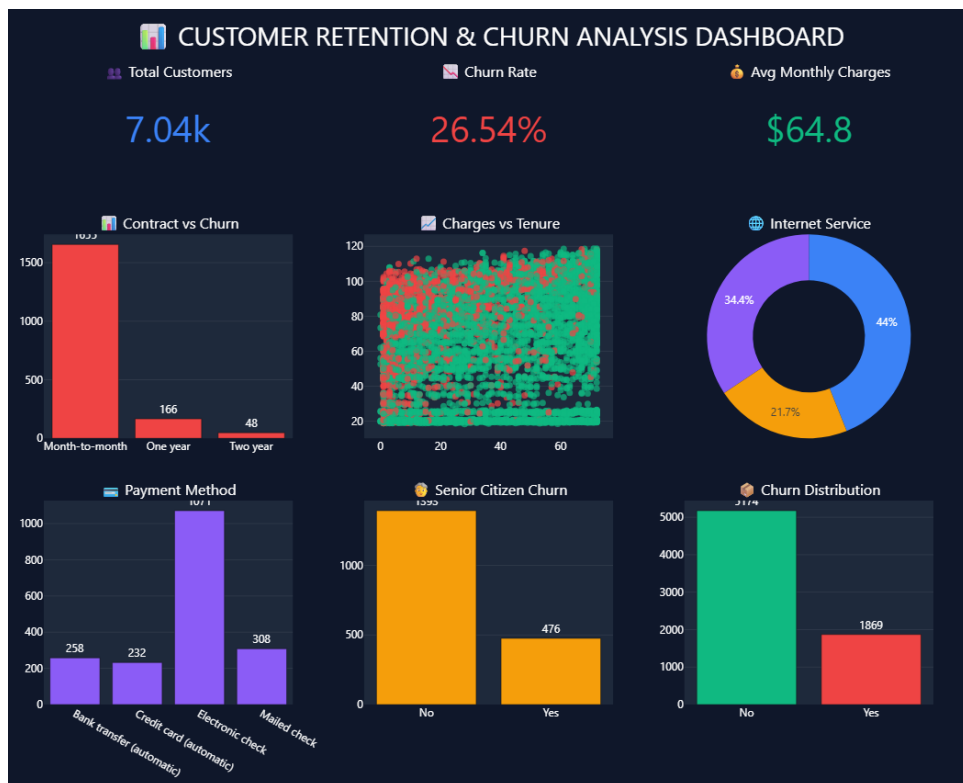


Customer Retention & Churn Analysis

Future Interns - Data Science & Analytics Internship

Task 2 Report



Submitted By:

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CIN ID: FIT/APR26/DS16984

Tools Used:

Power BI, Excel, Power Query

1. Abstract

This project focuses on analyzing customer churn behavior in a telecom subscription-based business using Python and data analytics techniques. The analysis aims to identify churn patterns, customer retention trends, service-related issues, and business factors influencing customer loss.

The project uses Exploratory Data Analysis (EDA), statistical visualization, and interactive dashboards to transform raw customer data into meaningful business insights. Various visualizations were developed using Matplotlib, Seaborn, and Plotly to understand customer behavior effectively.

The interactive dashboard highlights key metrics such as churn rate, customer segmentation, contract analysis, payment method analysis, and internet service distribution.

2. Introduction

Customer retention is one of the most critical factors for subscription-based businesses. Acquiring new customers is often more expensive than retaining existing ones. Therefore, analyzing customer churn behavior helps businesses improve customer satisfaction, optimize services, and reduce revenue loss.

This project presents a comprehensive churn analysis dashboard developed using Python visualization libraries. The dashboard provides insights into:

- Customer churn patterns
- Contract-based churn analysis
- Payment method impact
- Internet service distribution
- Monthly charges and customer tenure
- Customer retention behavior

The project demonstrates how data analytics can support data-driven business decision-making.

3. Objectives

The main objectives of this project are:

1. To analyze customer churn behavior.
2. To identify factors affecting customer retention.
3. To perform exploratory data analysis using Python.
4. To create interactive business dashboards.
5. To generate actionable business insights.
6. To provide recommendations for reducing customer churn.

4. Tools and Technologies Used

Tool	Purpose
Python	Data Analysis and Dashboard Development
Pandas	Data Cleaning and Preprocessing
NumPy	Numerical Operations
Matplotlib	Basic Data Visualization
Seaborn	Statistical Visualization
Plotly	Interactive Dashboard Creation
Jupyter Notebook	Development Environment

Table 1: Tools and Technologies Used

5. Dataset Description

The dataset contains telecom customer information including:

- Customer demographics
- Subscription details
- Internet service type
- Contract type
- Payment method
- Monthly charges
- Total charges
- Customer tenure
- Churn status

The dataset was cleaned and preprocessed before performing analysis.

6. Data Cleaning Process

The following preprocessing steps were performed using Pandas:

1. Handled missing values in the dataset.
2. Converted TotalCharges into numeric format.
3. Removed inconsistent values.
4. Standardized categorical variables.
5. Prepared data for visualization and analysis.

7. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand customer behavior and churn trends.

The analysis included:

- Univariate Analysis
- Bivariate Analysis
- Correlation Analysis
- Churn Distribution Analysis
- Customer Segmentation

Visualization libraries such as Matplotlib and Seaborn were used to create statistical plots and charts.

8. Dashboard Overview

The interactive dashboard was created using Plotly.

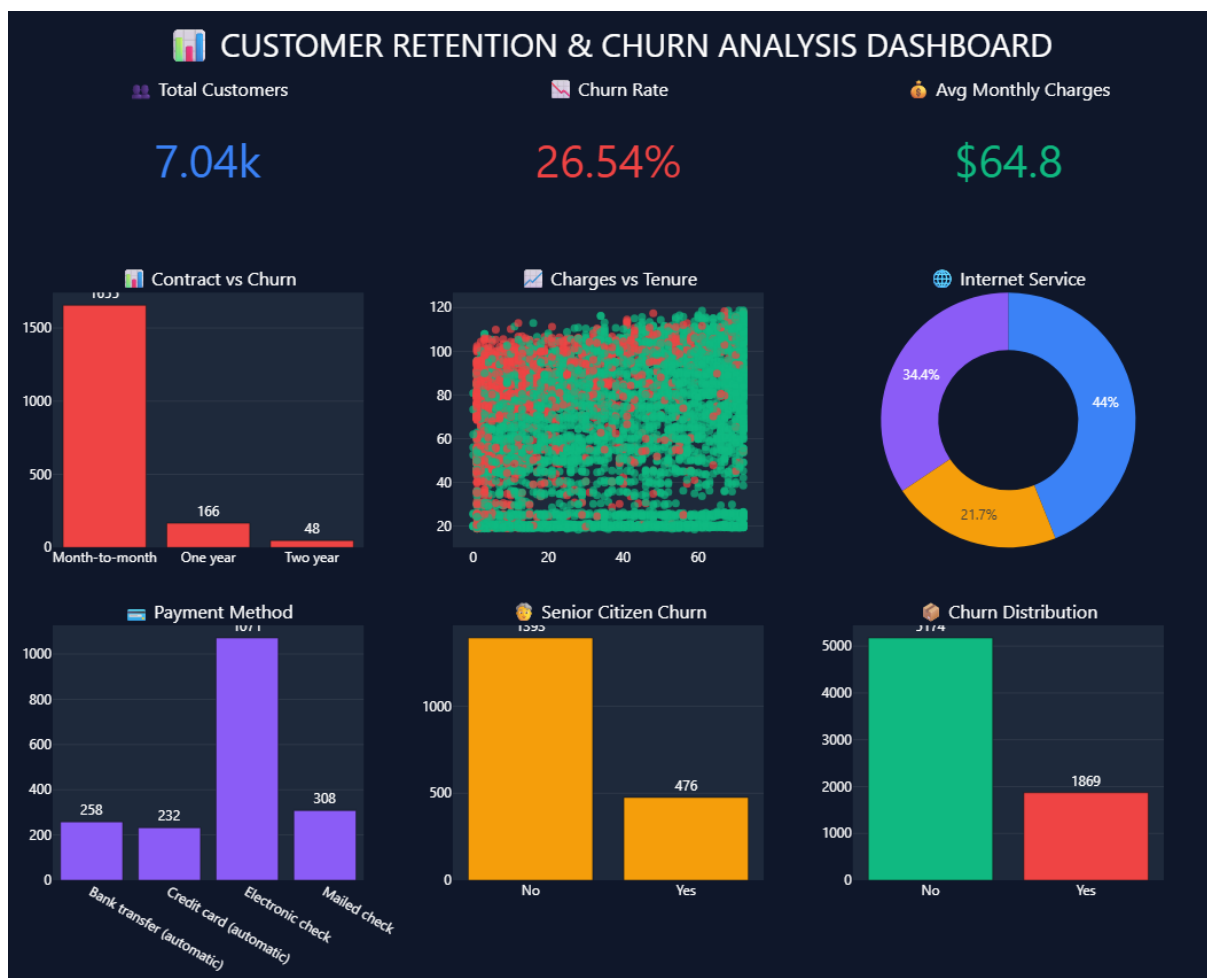


Figure 1: Customer Retention and Churn Dashboard

The dashboard includes:

- KPI Cards
- Churn Analysis
- Contract Type Analysis
- Payment Method Analysis
- Internet Service Distribution
- Monthly Charges vs Tenure Analysis
- Customer Segmentation Charts

9. KPI Analysis

KPI	Value
Total Customers	7043
Customer Churn Rate	26.54%
Average Monthly Charges	\$64.76
Retained Customers	5174
Churned Customers	1869

Table 2: Customer Churn KPIs

10. Key Insights

10.1 Month-to-Month Customers Have Highest Churn

Customers with month-to-month contracts showed significantly higher churn compared to long-term contract customers.

Business Impact: Businesses should encourage long-term subscriptions through discounts and loyalty programs.

10.2 Higher Monthly Charges Increase Churn Probability

Customers with higher monthly charges are more likely to leave the service.

Business Impact: Businesses should optimize pricing strategies and improve value-added services.

10.3 Fiber Optic Users Show Higher Churn

Fiber optic internet users contribute significantly to customer churn.

Business Impact: Improving service quality and customer support can improve retention.

10.4 Electronic Check Users Have Higher Churn

Customers using electronic check payment methods show higher churn behavior.

Business Impact: Encouraging automatic payment methods may improve customer retention.

10.5 Long-Term Customers Show Better Retention

Customers with higher tenure demonstrate stronger loyalty.

Business Impact: Businesses should focus on retaining new customers during early subscription periods.

11. Business Recommendations

Based on the analysis, the following recommendations are suggested:

1. Offer discounts for long-term contracts.
2. Improve customer support quality.
3. Launch targeted retention campaigns.
4. Optimize pricing strategies.
5. Encourage automatic payment methods.
6. Improve service quality for high-risk customers.

12. Challenges Faced

- Handling missing values and inconsistent data.
- Managing categorical variables.
- Designing balanced dashboard layouts.
- Selecting appropriate visualizations.
- Building interactive dashboards using Plotly.

13. Learnings

Through this project, the following skills were developed:

- Exploratory Data Analysis
- Dashboard Development
- Data Cleaning using Pandas
- Statistical Visualization
- Interactive Plotly Dashboard Design
- Business Insight Generation

14. Conclusion

This project successfully analyzed customer retention and churn behavior using Python and data visualization techniques.

The dashboard provides valuable insights into customer behavior, service-related issues, and churn patterns. The analysis can help businesses improve customer retention strategies, optimize services, and enhance customer satisfaction.

The project demonstrates the importance of business analytics and interactive dashboards in modern decision-making systems.

15. References

1. Python Documentation
2. Pandas Documentation
3. Matplotlib Documentation
4. Seaborn Documentation
5. Plotly Documentation
6. Telco Customer Churn Dataset
7. Future Interns Internship Guidelines